

Source and filter contributions to voice quality differences

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Abstract

This study examines how both the source and filter may contribute to voice quality differences. MRI and EGG recordings were obtained on a female subject who was trained in the Estill Voice method of voice production. The subject recorded two samples of the vowel /i/, at about 500 Hz, keeping larynx height constant, with two different fold masses. The subject varied vocal fold mass by producing a “thin” (falsetto) fold mass vs a “thick” (chest voice) quality. The ARX-LF model was used to estimate vocal tract area with the results indicating a larger oral cavity for the thick fold phonation than the thin fold one; a similar pattern was seen in the vocal tract midsagittal contours of the MRI images for the two /i/ vowels. In addition, F2 and F4 were lower for thick fold phonation than for thin. Open Quotient (OQ) was greater for thin than thick, with a corresponding steeper spectral tilt. Future work will examine more vowels and more speakers.

Keywords: source filter interaction, MRI, EGG, fold thickness, ARX-LF model

1. Introduction

This is an exploratory study to examine how the source and the filter contribute to voice quality differences. According to the source-filter theory of phonation (Fant 1970), the source and filter are for the most part independent. In more recent years researchers have begun to acknowledge source-filter interactions. For instance, supralaryngeal articulation can affect laryngeal behavior, (e.g., Rothenberg 1980; Fant and Lin 1987; Childers and Wong 1994; Titze 2004; Barney et al. 2007); and vice versa (e.g., Klatt & Klatt 1990; Barney et al. 2007; Erickson et al. 2020). Using modeling, such as the ARX-LF model (Li et al. 2018; Li, Sakakibara et al. 2019), provides a way to examine the interaction between source and filter. Erickson et al. (2019) and Li, Takahashi et al. (2019), using this model, reported how varying larynx height along with vocal fold thickness affects Open Quotient (OQ), formants, and spectral tilt. Another way yet to investigate source filter interactions is through singing studies, (e.g., Titze et al. 2003; Henrich et al. 2005; Miller & Schutte 2005; & Henrich-Bernadoni et al. 2014). Singers are generally able to control their laryngeal and supralaryngeal articulation to produce an intended voice quality/singing genre. Henrich et al. (2005) used EGG to examine how chest and falsetto modes of phonation

influenced glottal-source parameters such as the open/closed or contact quotient of the vocal folds. Chest and falsetto can also be referred to respectively as Mechanism 1 and Mechanism 2, and, as described in Henrich-Bernadoni et al. (2014) p. 491 “are thought to be associated, respectively, with the increase or decrease in vibrating mass that results from the coupling or decoupling (respectively) of the layered vocal fold structure to the inferior thyro-arytenoid or vocalis muscle (Hirano 1982). Within a certain range of frequencies lying between their low and high F0 range, a singer can phonate in either register (Vennard 1967; Roubeau et al. 2009). Acoustic differences for the two registers are that vowels produced in M2, compared to M1, have larger Open Quotient (OQ), steeper spectral tilt, and resonant formant frequencies also change (Henrich et al. 2005; Henrich-Bernadoni et al. 2014); specifically, they report that the first and second resonant frequencies of a vowel sung at the same F0 are higher in M2 compared to M1. These findings suggest that articulatory settings differ for M1 and M2 laryngeal modes. In their report, they speculate about what these supralaryngeal changes might be.

Echternach et al. (2010) used MRI recordings to examine 10 professional tenor voices as they sang a scale on the vowel /a/, progressing from chest range to falsetto range, but keeping their voice during the passaggio from low notes to high notes in what is called “voix mixte”, supposedly a mix between Mechanism 1 and Mechanism 2. They report the singers in this range between M1 and M2 tended to widen their pharyngeal areas, and increase their lip, jaw openings, and also jaw protrusion, thus creating a larger oral cavity.

In our study, we use EGG and MRI to examine the laryngeal and supralaryngeal differences in the production of two sustained /i/ vowels, produced on roughly the same F0, maintaining the larynx at mid position, but changing the thickness of the phonating vocal folds, along the lines of Hirano’s description. THIN vocal folds were produced with the upper edge and cover of the vocal folds, while THICK folds, with greater mass/body of the folds together both upper and lower medial edges. Roughly speaking, THIN folds correspond to falsetto-type phonation, and THICK folds, to modal-type (chest) phonation. Modal/chest phonation is often referred to as Mechanism 1, while falsetto, to Mechanism 2 (e.g., Henrich et al. 2004). Figure 1 shows a schematic illustration of the two phonation modes.

Singers trained in the method of voice production proposed by Estill et al. (2017) are purportedly able to control the thickness of their vocal folds while singing. The Estill method teaches singers as well as voice actors/actresses to control laryngeal and supralaryngeal articulation in order to produce voice qualities appropriate for the genre they want to sing, or in the case of voice actors/actresses, for the character they wish to portray. In everyday speech, voice quality changes, and people do this “automatically”, without knowing necessarily how they are changing their articulation, to appropriately express social attitudes (Rilliard et al. 2013) or ethnophonetic expressions (Sadanobu et al. 2016).

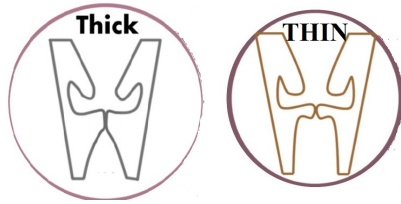


Figure 1: Schematic illustration of the two phonation modes: Thick mode involves vibration of both cover and body; thin, only the cover or upper edge of the vocal fold.

The two /i/ vowels studied in this paper were part of a larger MRI study (Erickson, Yun et al. 2020). For this experiment, a phonetician (first author) trained in the Estill method was the subject. Since these two vowels were produced by keeping F0 and larynx height constant, changing supposedly only the thickness of the vocal folds, these two /i/ vowels were chosen as candidates to examine source-filter interactions. Moreover, these two /i/ were part of an earlier perception study in which listeners were asked to judge voice quality differences (Erickson, Kawahara et al. 2020). The results showed that they rated the /i/ produced with thick folds as more aroused and more positive than that of the thin folds, indicating a perceptual change between the two /i/ vowels.

The questions we investigate here are (1) How does “thickness” of vocal folds affect Open Quotient (OQ), (2) Does “thickness” of vocal folds affect supralaryngeal settings, and (3) What are some acoustic changes associated with thick fold vs. thin fold mode of phonation? To examine some of the acoustic changes, we use the ARX-LF model (Li et al. 2018; Li, Sakakibara et al. 2019) for estimating source and filter contributions to the acoustic signal; also, we qualitatively compare the estimated area functions generated by the ARX-LF model with the vocal tract midsagittal contours of the MRI images.

2. Methods

2.1 MRI and EGG recordings

MRI and simultaneous acoustic recordings were made using the 3T MRI scanner (Siemens MAGNETOM Verio at the ATR Brain Activity Imaging Center (BAIC), Kyoto, Japan. For analysis of sustained articulator position and shape, static images of three sagittal slices (one for midsagittal slice and the others for right and left adjacent para-sagittal slices) were obtained by 2D FLASH sequences (TR=17ms, TE=1.62ms,

field of view= 256mm x 256mm, spatial resolution=128pixel x 128pixel, slice thickness=5mm, slice separation=5mm). A 1200 ms duration series of three 100 ms tone bursts whose onsets were at 0ms, 300ms and 600ms, (resulting in a perceptual pattern of da da DAH) was presented 128 times to the subject. The subject was asked to vocalize the target vowel sound repeatedly when the third noise burst (DAH) was presented and then sustain it until the first noise burst of the next cycle. A resting period was inserted every 10 repetitions to allow time to swallow and wet the mouth. The MRI scanner acquired 66 frames of MR images with a frame rate of 60fps for each of three slices by a trigger signal in sync with the first noise burst presented repeatedly to the subject.

For EGG recordings, vowels were recorded separately in a soundproof room at Arai Lab at Sophia University with an electroglottograph (Glottal Enterprises EG2-PCX2) and an electret condenser microphone (Sony ECM-MS957) connected to a laptop computer via an audio interface (Edirol UA-25EX). Acoustic and EGG signals were recorded simultaneously using Audacity at a 44.1 kHz sampling rate,

2.2. Corpus

The speaker, a female phonetician trained in the Estill Voice method (Estill et al. 2017), produced two sustained /i/ vowels, varying the thickness of the vocal folds (thick vs. thin). They were produced at approximately 500 Hz, and the larynx was kept at mid position.

2.3 Analysis

EGG analysis was done with Praatdet (Kirby 2017). OQ was estimated by detection of closing and opening peaks, similar to that reported in (Henrich et al. 2004; Kirby 2017) using vowel sounds recorded at Arai Lab, Sophia University. The acoustic analysis was done with the ARX-LF model (Li et al. 2018; Li, Sakakibara et al. 2019), using the two /i/ vowel sounds recorded during the MRI experiment.

3. Results

3.1 Source characteristics

Both the EGG and ARX-LF analysis showed that the /i/ produced with thin folds had larger Open Quotient (OQ) than that produced with thick folds. This is shown in Table 1. The first OQ value is that estimated by the ARX-LF model, and the second, that calculated from the EGG analysis. Although the two OQ estimates are not identical, they are in the same direction. The table also shows the spectral tilt estimates by the ARX-LF model. Figure 2 also shows the difference in spectral tilt. The tilt of the thin folds is much steeper than that of the thick folds, which is to be expected, given the smaller OQ for thin folds.

Table 1: ARX-LF model estimate of spectral tilt and OQ for /i/ phonated with thin and thick folds.

Phonation mode	Tilt	OQ _{model}	OQ _{egg}
thin (falsetto)	-15.6	0.47	0.78
thick (modal)	-12.7	0.4	0.55

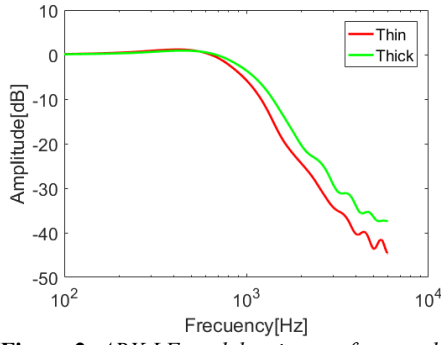


Figure 2: ARX-LF model estimate of spectral tilt.

3.2 Filter characteristics

The results of the MRI imaging are shown in Fig. 3. When we compare the thin folds (left image) with the thick folds (middle image), we see that even though the speaker maintained the relatively same F0 and the relatively same mid larynx position for the two productions, the vocal tract midsagittal contours are different. For the thick fold phonation, the vocal tract area appears larger: the mouth is more open, the tongue is more raised, fronted and bunched, leaving the oropharynx larger, and also the posterior oral cavity larger.



Figure 3: MRI images of two /i/ vowels. From left to right thin vocal folds, thick vocal folds, overlay of thin (pink) and thick (green). For the overlay, pink (thin folds) were moved upward and forward, and green (thick folds) were moved backward and downward.

3.3 ARX-LF model estimate of vocal tract area

The estimated area functions as calculated from the ARX-LF model are shown in Figure 4. We see an overall larger area for the /i/ produced with thick folds than with thin folds. Specifically, we see an increase in 3 regions: from lips to 3, from 4 to 7 and from 7 to 14. These seem to correspond well to the larger areas of the MRI image for the thick folds.

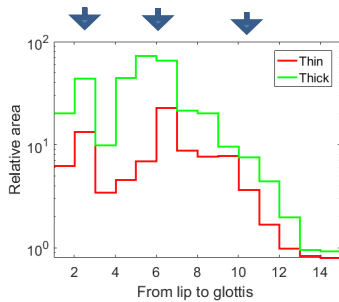


Figure 4: ARX-LF model estimates of area functions. Y-axis indicates relative area, x-axis relative length from lip (left) to glottis (right). Estimated area for thick folds marked in green, for thin folds, in red.

As a result of differences in vocal tract configurations, we see different formant resonance frequencies of the vocal tract. Figure 5 shows the ARX-LF model estimate of the formants, and Table 2 shows the actual values. Based on this analysis we see that F2 and F4 are lower for the /i/ produced with thick folds than that produced with thin folds. This is also shown clearly in Table 2.

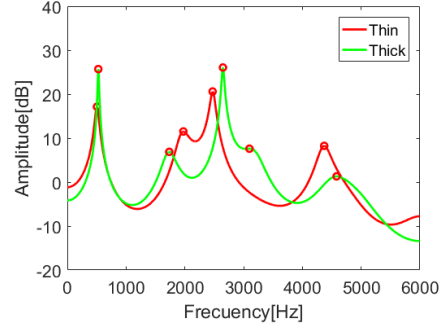


Figure 5: ARX-LF model estimate of formants. Y-axis indicates amplitude in dB, x-axis, frequency. Formants for thick folds in green, for thin folds, in red.

Table 2: ARX-LF model estimate of F0 and formants for /i/ phonated with thin and thick folds

Phonation mode	F0	F1	F2	F3	F4
thin (falsetto)	500	510	1980	2479	4377
thick (modal)	520	533	1740	2654	3105

4. Discussion and conclusion

In summary, we found interactions between source and filter: both the source and the filter characteristics change when vocal fold thickness changes. Specifically, /i/ produced with thick folds, compared to thin folds, has smaller OQ and less steep spectral tilt. In addition, we also see supralaryngeal articulation changes due to mode of phonation. For thick fold phonation, the oral cavity increases consistently throughout the vocal tract (front, mid and back). Formants also change: lower F2 and F4 for thick fold compared to thin fold phonation.

Previous studies have also reported larger OQ and steeper spectral tilt for thin folds (falsetto) than thick folds (chest). About formant differences, Henrich-Bernadoni et al. (2014) also reported formant differences between falsetto and chest voice, speculating concomitant changes in mouth opening or tongue shape; specifically, for M2 (thin folds), there may be a “small but measurable decrease in jaw opening expected to produce a decrease in R1” [p.499]. Congruent to this speculation, our findings show less jaw opening for thin folds. Changes in supraglottal articulation for different phonation modes, similar to what we found, was also reported by Echternacht et al. (2010).

However, with regard to formants, Henrich-Bernadoni et al. (2014) reported higher frequencies for R1 and R2 for falsetto (i.e., thin folds), whereas we report higher F2 and F4, but not for F1. This may be because Henrich-Bernadoni et al. (2014)

examined low /a/ vowels, while we examined high /i/ vowels. In our study with /i/, we found that for thick folds, F1 is higher and F2 is lower, thus bringing F1 and F2 closer together. This finding is consistent with findings by Story et al. (2001) that a wide oral cavity and increased tract length may cause F1 and F2 to be closer together. However, in our data, we show only increased oral/pharyngeal areas, no vocal tract lengthening, as the speaker was purposely keeping the larynx at mid position for the two phonation modes. Enlarging the vocal tract when phonating with thicker folds, instead of lowering the larynx, may be in order to help maintain phonation, similar to what happens when enlarging the supraglottal tract during phonation of voiced stops, as reported by e.g., Ohala (1983).

In summary, an interesting finding in our study is that supralaryngeal gestures indeed change when phonation mode is changed—the vocal tract is enlarged for the thick fold phonation vs. thin fold. Specifically, for this speaker who maintained vocal tract length, she opened the mouth more, and raised, fronted and bunched the tongue more, thus leaving the oropharynx larger, and also the posterior oral cavity larger. The ARX-LF model estimated area functions seem to agree with the midsagittal vocal tract contours from the MRI images. Future work will focus on quantitative analyses of the MRI images of the vocal tract areas for comparison with estimated acoustic outputs generated by the ARX-LF model. More vowels and more speakers will also be examined.

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